Saif Mostasir ROHAN

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264/1, Chowrasta, Gazipur-1702, Bangladesh

Computer Science and Engineering graduate from American International University-Bangladesh (AIUB), currently pursuing an MSc in Computer Science and Engineering with a focus on Intelligent Systems. I am focused on AI engineering, with hands-on work in LLM-powered applications, RAG, agentic workflows, vector databases, backend APIs, observability, and deployment. My earlier background in AI research, computer vision, explainable AI, and peer-reviewed publications helps me connect research ideas with practical, production-aware AI systems.

SKILLS

Programming Languages	Python, JavaScript, SQL, C, C++, C#, R, MATLAB, HTML, PHP
AI Engineering	LLM Applications, RAG, Agentic AI, Tool Calling, Prompt Engineering, Human-in-the-Loop Workflows, Research Automation
Frameworks & Libraries	LangChain, LangGraph, FastAPI, Streamlit, TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas
Vector Search & Data	ChromaDB, Embeddings, Vector Stores, Semantic Search, Document Retrieval, Knowledge Bases, SQLite
Deployment & Backend	FastAPI, REST APIs, Docker, Hugging Face Spaces, Git, GitHub, API Integration, Backend Services
Observability & Reliability	LangSmith, Prometheus, Logging, Evaluation, Retry Handling, Circuit Breakers, Production-Aware AI Systems
Computer Vision & Research	Computer Vision, Image Classification, Image Segmentation, Explainable AI, Multimodal Learning
Tools & Platforms	Google Colaboratory, Kaggle, LaTeX, Microsoft Office 365, Figma, Canva, Draw.io, Windows, macOS, Linux Ubuntu

PROJECTS

PROJECT AEGIS : AUTONOMOUS MEDICAL RESEARCH ASSISTANT — 2026

github.com/saifmostasirrohan/project-aegis-phase   Live Demo on Hugging Face

Built and deployed an autonomous AI research assistant for medical literature review and research synthesis. The system accepts a research query, retrieves scientific papers, compares external findings with a local research knowledge base, runs LLM-powered reasoning locally, pauses for human-in-the-loop approval, and generates a structured literature review with summaries, comparison tables, and future research directions. The project demonstrates agentic AI orchestration, RAG, vector memory, tool integration, observability, and production-aware reliability patterns.

Python · Local LLM · LangGraph · LangChain · MCP · FastAPI · Streamlit · ChromaDB · SQLite · LangSmith · Prometheus · Docker · Hugging Face Spaces

EXPERIENCE

Summer 2024–2025

Teaching Assistant, AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB), Dhaka, Bangladesh

- Assisted academic activities under the supervision of Prof. Dr. Afroza Nahar.
- Supported students in course-related learning, academic discussion, and technical understanding.
- Contributed to classroom and laboratory support activities related to computer science coursework.

Teaching · Academic Support · Computer Science · Student Mentoring

EDUCATION

Jan 2026 – Ongoing	**MSc in Computer Science and Engineering**, Major in Intelligent Systems, American International University-Bangladesh (AIUB), Dhaka, Bangladesh
2024 – 2025	**BSc in Computer Science and Engineering**, Major in Information Systems, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. Graduate : Summer 2024–2025. Total Credits : 148. CGPA : 3.91/4.00
2020	**Higher Secondary Certificate (HSC)**, Science, M. E. H. Arif College, Gazipur, Bangladesh. GPA : 5.00/5.00
2018	**Secondary School Certificate (SSC)**, Science, Rani Bilasmoni Govt. Boys' High School Joydebpur, Gazipur, Bangladesh. GPA : 5.00/5.00

Peer-Reviewed Publications

1. **S. M. Rohan**, S. S. Rafith, M. S. Kabir, M. H. Pranto, T. Bhuiyan, and K. Tanvir, “XGception : Unveiling Mosquito Larvae Patterns with XAI,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 2025. **[Link]**
2. S. S. Rafith, **S. M. Rohan**, M. S. Kabir, T. S. Sintheia, N. S. B. Alam, and K. Tanvir, “Explainable Wildfire Classification with PVT and Dragonfly-Optimized Decision Trees,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 2025. **[Link]**
3. S. S. Rafith, **S. M. Rohan**, M. S. Kabir, K. Tanvir, D. Gomes, and M. Rahman, “Explainable Breast Cancer Diagnosis with Vision Transformers and ACOR-Optimized Decision Trees,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 2025. **[Link]**
4. S. S. Rafith, **S. M. Rohan**, M. S. Kabir, M. F. K. Chowdhury, and A. Nahar, “FractureNet : An Optimized Hybrid Transformer-Ensemble Architecture for Bone Fracture Detection in Multi-Region X-ray Images,” *International Conference of Frontiers of Engineering and Emerging Technologies (FET’26)*, 2026. **[Accepted, IEEE]**
5. **S. M. Rohan**, T. I. Sheikh, F. B. Zaman, M. Ahmed, M. S. Kabir, and K. T. Hasan, ”ReedMap : A Tangible Interaction System for Context-Aware and Division-Based Map Navigation,” in *Proceedings of the 28th International Conference on Human-Computer Interaction (HCII)*, Montreal, Canada, 2026. **[Accepted, Springer LNCS]**
6. T. I. Sheikh, **S. M. Rohan**, N. Aziz, S. F. Mahbub, A. Amin, and K. T. Hasan, ”AirTune : A Tangible Interaction System for Gesture-Based and Contactless Media Control,” in *Proceedings of the 28th International Conference on Human-Computer Interaction (HCII)*, Montreal, Canada, 2026. **[Accepted, Springer LNCS]**
7. A. D. Sarker, **S. M. Rohan**, M. F. K. Chowdhury, S. S. Rafith, N. Fahim, and M. S. Kabir, “Capsule Endoscopy Image Classification using ResNeXt & VCS-Optimized KNN with XAI,” *2025 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, Rajshahi, Bangladesh, 2025. **[Link]**
8. N. Fahim, S. S. Rafith, M. F. K. Chowdhury, **S. M. Rohan**, A. D. Sarker, and M. S. Kabir, “Colonoscopy Classification using PVT-Xception Ensemble Framework with Explainable AI,” *2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2025. **[Link]**
9. T. S. Sintheia, M. Rahman, B. Sutrudhar, **S. M. Rohan**, M. M. Rassel, and M. S. Kabir, “Blood Cancer Classification from Microscopic Images via ViTResNetEnsemble with XAI,” *2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, Dhaka, Bangladesh, 2025. **[Link]**

Language Skills

- **Bengali :** Native Fluency
- **English :** Native Fluency
- **Academic IELTS :** Overall Band 7.5
- **IELTS Component Scores :** Listening 7.5, Reading 8.5, Writing 6.5, Speaking 6.5

References

1. **Prof. Dr. Khandaker Tabin Hasan**
   Associate Dean, Faculty of Science and Technology
   Department of Computer Science
   American International University-Bangladesh (AIUB), Dhaka, Bangladesh
   Email : tabin@aiub.edu

2. **Prof. Dr. Afroza Nahar**
   Professor, Faculty of Science and Technology
   Department of Computer Science
   American International University-Bangladesh (AIUB), Dhaka, Bangladesh
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